



RV College of Engineering®

Mysore Road, RV Vidyaniketan Post, Bengaluru - 560059, Karnataka, India

Go, change the world

Federated Heterogeneous Graph Neural Networks for TBML Detection

A Major Project Report (CS481P)

Submitted by,

Nikhil Vasu

1RV22CS128

Rohith Biradar

1RV22CS164

Suraj Chanaveeragoudra

1RV22CS211

Under the guidance of

Dr. Nagaraj G. S

Prof. Sanjana Ravindra Otihal

Professor

Assistant Professor

Dept. of CSE

Dept. of CSE

RV College of Engineering

RV College of Engineering

In partial fulfillment of the requirements for the degree of

Bachelor of Engineering in

Computer Science and Engineering

2025-26

RV College of Engineering®, Bengaluru
(Autonomous institution affiliated to VTU, Belagavi)
Department of Computer Science and Engineering



CERTIFICATE

Certified that the major project (CS481P) work titled ***Federated Heterogeneous Graph Neural Networks for TBML Detection*** is carried out by **Nikhil Vasu (1RV22CS128)**, **Rohith Biradar (1RV22CS164)** and **Suraj Chanaveeragoudra (1RV22CS211)** who are bonafide students of RV College of Engineering, Bengaluru, in partial fulfillment of the requirements for the degree of **Bachelor of Engineering in Computer Science and Engineering** of the Visvesvaraya Technological University, Belagavi during the year 2025-26. It is certified that all corrections/suggestions indicated for the Internal Assessment have been incorporated in the major project report deposited in the departmental library. The major project report has been approved as it satisfies the academic requirements in respect of major project work prescribed by the institution for the said degree.

Guide

Head of the Department

Principal

Dr. Nagaraj G. S

Dr. Shanta Rangaswamy

Dr. K. N. Subramanya

External Viva

Name of Examiners

Signature with Date

1.

2.

DECLARATION

We, **Nikhil Vasu, Rohith Biradar** and **Suraj Chanaveeragoudra** students of eighth semester B.E., Department of Computer Science and Engineering, RV College of Engineering, Bengaluru, hereby declare that the major project titled '**Federated Heterogeneous Graph Neural Networks for TBML Detection**' has been carried out by us and submitted in partial fulfilment for the award of degree of **Bachelor of Engineering in Computer Science and Engineering** during the year 2025-26.

Further we declare that the content of the dissertation has not been submitted previously by anybody for the award of any degree or diploma to any other university.

We also declare that any Intellectual Property Rights generated out of this project carried out at RVCE will be the property of RV College of Engineering, Bengaluru and We will be one of the authors of the same.

Place: Bengaluru

Date:

Name

Signature

1. Nikhil Vasu(1RV22CS128)
2. Rohith Biradar(1RV22CS164)
3. Suraj Chanaveeragoudra(1RV22CS211)

ACKNOWLEDGEMENTS

We are indebted to our guide, **Dr. Nagaraj G. S**, Professor, Department of CSE, RVCE for the wholehearted support, suggestions and invaluable advice throughout our project work and also helped in the preparation of this thesis.

We also express our gratitude to our panel members **Dr. Nagaraj G. S**, Professor, Department of CSE, RVCE and **Prof. Sanjana Ravindra Otihal**, Assistant Professor, Department of CSE, RVCE for the valuable comments and suggestions during the phase evaluations.

Our sincere thanks to the project coordinators **Dr. Chethana R Murthy**, Professor, Department of CSE and **Prof. Deepika Dash**, Assistant Professor, for their timely instructions and support in coordinating the project.

Our gratitude to **Prof. Narashimaraja P**, Department of ECE, RVCE for the organized latex template which made report writing easy and interesting.

Our sincere thanks to **Dr. Shanta Rangaswamy**, Professor and Head, Department of Computer Science and Engineering, RVCE for the support and encouragement.

We express sincere gratitude to our beloved Professor and Vice Principal, **Dr. Geetha K S**, RVCE and Principal, **Dr. K. N. Subramanya**, RVCE for the appreciation towards this project work.

We thank all the teaching staff and technical staff of Computer Science and Engineering department, RVCE for their help.

Lastly, we take this opportunity to thank our family members and friends who provided all the backup support throughout the project work.

ABSTRACT

Trade-Based Money Laundering (TBML) has become a major challenge in modern financial systems due to the increasing complexity and scale of global trade transactions. Conventional anti-money laundering systems primarily rely on rule-based monitoring and traditional machine learning techniques, which are often inadequate for identifying hidden transactional dependencies and evolving laundering strategies. These approaches suffer from limitations such as high false positive rates, poor relational learning capability, and reduced scalability in large financial networks. To address these issues, this report presents a Federated Heterogeneous Graph Neural Network (HGNN) framework for real-time TBML detection capable of modeling complex trade relationships while preserving institutional data privacy.

The primary objective of this work is to develop a scalable and privacy-preserving TBML detection architecture using graph-based deep learning methodologies. The proposed framework represents financial ecosystems as heterogeneous graphs consisting of entities such as accounts, importers, exporters, transactions, and trade documents. Graph Attention Network (GAT)-based convolutional layers are employed to learn structural and semantic relationships between interconnected entities for effective anomaly detection. Federated learning mechanisms are incorporated to facilitate collaborative model training across distributed financial institutions without centralized data sharing. The computational framework includes graph construction, feature extraction, node embedding generation, suspicious activity classification, and real-time risk score prediction.

The proposed system is implemented using Python, PyTorch Geometric, FastAPI, Memgraph, PostgreSQL, and React.js. Experimental evaluation is carried out on simulated TBML datasets containing both legitimate and suspicious trade transactions. Performance analysis is conducted using standard classification metrics including accuracy, precision, recall, and F1-score. The experimental results indicate that the proposed framework effectively captures hidden laundering patterns and improves fraud detection capability when compared with conventional machine learning approaches. The architecture further supports scalable deployment, reduced false positive rates, and efficient real-time monitoring of high-volume trade transactions.

CONTENTS

Abstract	i
List of Figures	iv
List of Tables	v
1 Introduction to Trade-Based Money Laundering	1
1.1 Overview of Trade-Based Money Laundering	2
1.2 Literature Review	4
1.2.1 Sample	5
1.3 Motivation	6
1.4 Problem statement	6
1.5 Objectives	7
1.6 Brief Methodology of the project	7
1.7 Assumptions made / Constraints of the project	7
1.8 Organization of the report	7
2 Theory and Fundamentals of Analog to Digital converter	8
2.1 Contents of this chapter	9
2.2 Contents of this chapter	9
2.3 Contents of this chapter	9
2.4 Contents of this chapter	9
2.5 Use of Acronyms and Glossaries	9
3 Analysis/Design of Pipelined Analog to Digital converter	11
3.1 Contents of this Chapter	12
3.2 Paraphrasing	12
3.3 Quotations	13
4 Implementation of Pipelined Analog to Digital converter	14
4.1 Contents of this chapter	15

5	Results & Discussions	16
5.1	Contents of this chapter	17
5.2	Tables in thesis	17
5.3	Math equation in thesis	18
5.4	Interpretation of results	19
6	Conclusion and Future Scope	20
6.1	Conclusion	21
6.2	Future Scope	21
6.3	Learning Outcomes of the Project	21
A	Code	22
A.1	First Appendix	23
	Bibliography	24



LIST OF FIGURES



LIST OF TABLES

1.1	Key Statistics Related to Global Trade-Based Financial Crime	3
5.1	Country List	18



The logo of RV College of Engineering is a circular emblem. It features a central shield divided vertically, with a large 'R' on the left and a large 'V' on the right. The shield is encircled by a ring containing the text 'RASHTREEYA SIKSHANA SAMITHI TRUST' at the top and 'INSTITUTIONS' at the bottom. A small registered trademark symbol (®) is located to the upper right of the emblem.

Chapter 1

Introduction to Trade-Based Money

Laundering

CHAPTER 1

INTRODUCTION TO TRADE-BASED MONEY LAUNDERING

The rapid growth of international trade and digital financial infrastructure has significantly increased the complexity of monitoring global financial transactions. Alongside economic globalization, financial institutions are facing increasingly sophisticated forms of illicit financial activities that exploit weaknesses in conventional Anti-Money Laundering (AML) systems. Among these, Trade-Based Money Laundering (TBML) has emerged as one of the most challenging financial crimes due to its ability to conceal illegal fund transfers within legitimate trade operations. The distributed and interconnected nature of modern trade ecosystems necessitates intelligent, scalable, and privacy-preserving analytical frameworks capable of identifying hidden transactional relationships in real time.

“For financial institutions, TBML is uniquely challenging because illicit funds are embedded within legitimate trade transactions, leaving traditional AML controls blind to the risks.”

– Dr. Sebastian Hetzler, IMTF Co-CEO [1]

As highlighted in the above statement, Trade-Based Money Laundering exploits the complexity of international commerce by embedding illicit financial flows within apparently legitimate trade transactions. This significantly limits the effectiveness of conventional compliance systems that primarily rely on isolated transaction monitoring and static rule-based verification mechanisms.

1.1 Overview of Trade-Based Money Laundering

Trade-Based Money Laundering (TBML) refers to the process of disguising the proceeds of crime and transferring value through legitimate international trade transactions in an attempt to legitimize illicit financial activities [2]. Criminal organizations exploit weaknesses in global trade systems using techniques such as over-invoicing, under-invoicing, phantom shipments, duplicate invoicing, and falsification of trade documentation. Since these activities are embedded within normal commercial operations involving multiple intermediaries, jurisdictions, and financial entities, identifying suspicious

transactional behavior becomes highly challenging for regulatory agencies and financial institutions.

The scale and economic impact of money laundering have increased considerably with the expansion of global commerce and digital financial ecosystems. According to the United Nations Office on Drugs and Crime (UNODC), money laundering accounts for nearly 2% to 5% of the global Gross Domestic Product (GDP), corresponding to approximately \$800 billion to \$2 trillion annually [3]. Furthermore, Europol reports that despite extensive regulatory frameworks and compliance procedures, global authorities successfully intercept less than 1% of illicit financial flows [4]. These statistics highlight the growing sophistication of modern laundering networks and the limitations of existing monitoring infrastructures.

As financial institutions adopt stricter Know-Your-Customer (KYC) regulations and transaction surveillance systems, criminal organizations increasingly rely on TBML mechanisms to avoid direct cash movement and bypass conventional banking controls. Reports published by Global Financial Integrity indicate that trade misinvoicing and related TBML activities account for nearly 87% of illicit financial flows originating from developing economies [5]. Similarly, industry analyses estimate that approximately \$1.6 trillion to \$2 trillion in illicit value is concealed annually within legitimate trade invoices and commercial documentation [1]. These findings demonstrate the critical role of international trade channels in enabling large-scale financial crime.

Table 1.1: Key Statistics Related to Global Trade-Based Financial Crime

Financial Crime Metric	Estimated Value
Global GDP associated with money laundering annually	2% – 5%
Estimated annual TBML economic volume	\$1.6T – \$2.0T
Illicit flows linked to trade misinvoicing in developing economies	87%
Global interception rate of illicit financial flows	<1%
Reduction in false positive alerts using AI-driven monitoring systems	Up to 40%

Conventional Anti-Money Laundering systems primarily depend on rule-based monitoring engines, predefined transaction thresholds, and manual auditing processes. AI-

though these approaches are capable of identifying isolated suspicious transactions, they often fail to capture hidden relational dependencies among entities involved in coordinated laundering operations. Modern TBML networks operate through interconnected ecosystems involving importers, exporters, logistics providers, shell companies, offshore entities, and financial intermediaries distributed across multiple jurisdictions. Since traditional machine learning models analyze transactions independently, they are unable to effectively model structural relationships and behavioral interactions embedded within large-scale financial networks.

Recent advancements in Graph Neural Networks (GNNs) and federated learning have introduced new opportunities for intelligent financial fraud detection. Graph-based learning techniques enable financial ecosystems to be represented as interconnected graphs where entities such as accounts, organizations, transactions, and trade documents are modeled as nodes and edges. Such graph representations allow deep learning systems to capture hidden transactional patterns and anomalous structural relationships that are difficult to identify using conventional machine learning approaches. In addition, federated learning facilitates collaborative model training across distributed financial institutions without requiring centralized sharing of sensitive customer data, thereby ensuring privacy preservation and regulatory compliance.

This project proposes a Federated Heterogeneous Graph Neural Network (HGNN) framework for real-time detection of Trade-Based Money Laundering activities. The proposed system integrates heterogeneous graph construction, Graph Attention Network (GAT)-based learning, federated intelligence, and real-time analytics to improve fraud detection capability in large-scale financial environments. The framework leverages technologies such as PyTorch Geometric, Memgraph, FastAPI, PostgreSQL, and React.js to develop a scalable and privacy-preserving AML platform capable of identifying suspicious transactional behavior across distributed trade ecosystems.

1.2 Literature Review

A literature review is a text of a scholarly paper, which includes the current knowledge including substantive findings, as well as theoretical and methodological contributions to a particular topic. Literature reviews are secondary sources, and do not report new or original experimental work. Most often associated with academic-oriented literature, such reviews are found in academic journals, and are not to be confused with book reviews

that may also appear in the same publication. Literature reviews are a basis for research in nearly every academic field . A narrow-scope literature review may be included as part of a peer-reviewed journal article presenting new research, serving to situate the current study within the body of the relevant literature and to provide context for the reader. In such a case, the review usually precedes the methodology and results sections of the work.

1.2.1 Sample

The main types of literature reviews are: evaluative, exploratory, and instrumental. A fourth type, the systematic review, is often classified separately, but is essentially a literature review focused on a research question, trying to identify, appraise, select and synthesize all high-quality research evidence and arguments relevant to that question. A meta-analysis is typically a systematic review using statistical methods to effectively combine the data used on all selected studies to produce a more reliable result.

Review types

The main types of literature reviews are: evaluative, exploratory, and instrumental. A fourth type, the systematic review, is often classified separately, but is essentially a literature review focused on a research question, trying to identify, appraise, select and synthesize all high-quality research evidence and arguments relevant to that question. A meta-analysis is typically a systematic review using statistical methods to effectively combine the data used on all selected studies to produce a more reliable result.

Process and product

Distinguish between the process of reviewing the literature and a finished work or product known as a literature review. The process of reviewing the literature is often ongoing and informs many aspects of the empirical research project. All of the latest literature should inform a research project. Scholars need to be scanning the literature long after a formal literature review product appears to be completed.

Page limitation

A careful literature review is usually 15 to 30 pages and could be longer. The process of reviewing the literature requires different kinds of activities and ways of thinking and link the activities of doing a literature review with Benjamin Bloom's revised taxonomy of the cognitive domain (ways of thinking: remembering, understanding, applying, analysing,

evaluating, and creating).

This section should contain the review of the literature in the past. You should review a minimum of 10 papers from standard reference journals. Kindly avoid local conference papers and papers from predatory journals. Kindly consult with your guide and finalize papers to be considered for review before adding in this section. Report the major observations and findings from each paper in one paragraph in the format given below.

proposed various techniques for adders and multipliers. Add the reference papers to the bibliography section using Jabref and cite it here using the instructions given in further chapters.

Plagiarism

To use someone else's exact words without quotation marks and appropriate credit, or to use the unique ideas of someone else without acknowledgement, is known as plagiarism. In publishing, plagiarism is illegal; in other circumstances, it is, at the least, unethical. You may quote or paraphrase the words or ideas of another if you document your source. Although you need not enclose the paraphrased material in quotation marks, you must document the source.

Paraphrased ideas are taken from someone else whether or not the words are identical. Paraphrasing a passage without citing the source is permissible only when the information paraphrased is common knowledge in a field. (Common knowledge refers to historical, scientific, geographical, technical, and other type of information on a topic readily available in handbooks, manuals, atlases and other references).

How to add Reference

Use Jabref which will help in adding the reference in a separate file, from which one can use `\citep{\}` command to add reference. A sample, referring to a textbook would look something like this,[6].

1.3 Motivation

Brief the motivation of selecting your project title. You can elaborate the challenges in the specific area, relevance and importance of the chosen topic.

1.4 Problem statement

Define the problem statement in this section, in one paragraph.

1.5 Objectives

The objectives of the project are

1. To design a pipelined ADC for audio frequency range
2. List all the objectives in the above format , starting with "To"
3. Limit the number of objectives to a maximum of three

1.6 Brief Methodology of the project

Discuss about the methodology you identified to execute the objectives of your project in brief. Methodology is a system of practices, techniques, procedures, and rules used to execute a particular project. You can elaborate the methodology in a later chapter. Here you can present in the form of a flow diagram and explain the methodology in a paragraph.

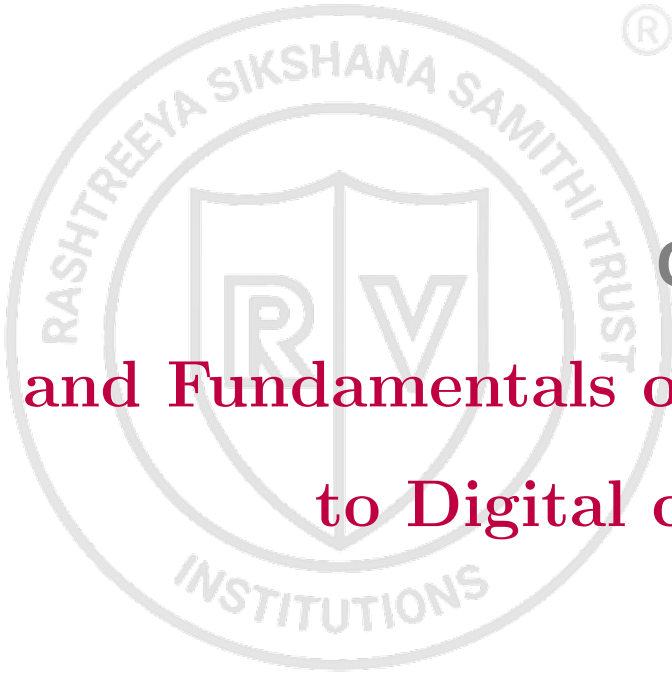
1.7 Assumptions made / Constraints of the project

List the assumptions made for the execution of the project in this section. You can also elaborate on the major constraints of the project. This section should clearly state under what conditions your project is valid. It is mandatory to have this section in your project report.

1.8 Organization of the report

This report is organized as follows. Write the discussions in each chapter. A sample is as follows.

- Chapter 2 discusses the fundamentals of ADC and the performance parameters for evaluation.
- Chapter 3 discusses .
- Chapter 4 discusses .
- Chapter 5 discusses .
- Chapter 6 discusses .

The logo is a circular emblem. The outer ring contains the text 'RASHTREEYA SIKSHANA SAMITHI TRUST' at the top and 'INSTITUTIONS' at the bottom. In the center is a shield divided vertically, with a large 'R' on the left and a large 'V' on the right.

Chapter 2

Theory and Fundamentals of Analog to Digital converter

CHAPTER 2

THEORY AND FUNDAMENTALS OF ANALOG TO DIGITAL CONVERTER

Every chapter should start with an introduction paragraph. This paragraph should brief about the flow of the chapter. This introduction can be limited within 4 to 5 sentences. The chapter heading should be appropriately modified (a sample heading is shown for this chapter). But don't start the introduction paragraph in the chapters 2 to end with "This chapter deals with....". Instead you should bring in the highlights of the chapter in the introduction paragraph.

2.1 Contents of this chapter

This chapter should discuss about the prerequisite learnings before the execution of the project. Organise and elaborate the theory and necessary fundamentals required for the execution of the project. You can use \subsections and subsubsections in this chapter.

2.2 Contents of this chapter

If a specific programming language is required for the project, a section can be allotted in this chapter to discuss it.

2.3 Contents of this chapter

Tools used could be another possible section to discuss about the software tools used in the work.

2.4 Contents of this chapter

The details in this chapter can be added in consultation with the project guide. For an internship based projects, subsections can be modified accordingly.

2.5 Use of Acronyms and Glossaries

Acronyms are nothing but the short form of regular repeated word. Say for example, you have a repeat word "Integrated Circuits" and you want to use a short form for it as "IC". For which you have to first define the word and use it wherever you wanted to refer it.

First, let's look at the definition, which has to be entered in `Glossaries.tex` under `CoverPages` directory.

```
%\newacronym{<Ref>}{<Short-Form>}{<Expanded word>}  
\newacronym{ic}{IC}{Integrated Circuits}
```

In order to use the defined acronym, use the commands `\gls{<Ref>}` as shown below

As an example, call the definition with `\gls{ic}` and the outcome of it is reflected as, Integrated Circuit (IC).

Note: For the First time, the expanded form appears along with the Short-form definition inside parenthesis. But when the `\gls{}` is repeated, only Short-form appears inside the parenthesis.

Now, let's look at the definition of symbols. Follow the syntax to define the symbol first, inside `Glossaries.tex` under `CoverPages` directory.

```
%\newglossaryentry{<Ref>}{name=<Symbol>, description={<description about the symbol>}}  
\newglossaryentry{rc}{name=$\tau$, description={Time constant}, type=symbolList}
```

As an example, the rate of change is defined with `\gls{rc}` and the outcome of it is reflected as, the rate of change is defined with τ .

The chapters should not end with figures, instead bring the paragraph explaining about the figure at the end followed by a summary paragraph.

After elaborating the various sections of the chapter (From Chapter 2 onwards), a summary paragraph should be written discussing the highlights of that particular chapter. This summary paragraph should not be numbered separately. This paragraph should connect the present chapter to the next chapter.

Chapter 3
**Analysis/Design of Pipelined Analog
to Digital converter**

CHAPTER 3

ANALYSIS/DESIGN OF PIPELINED ANALOG TO DIGITAL CONVERTER

Every chapter should start with an introduction paragraph. This paragraph should brief about the flow of the chapter. This introduction can be limited within 4 to 5 sentences. The chapter heading should be appropriately modified (a sample heading is shown for this chapter). But don't start the introduction paragraph in the chapters like "This chapter deals with....". Instead you should bring in the highlights of the chapter in the introduction paragraph.

3.1 Contents of this Chapter

This chapter should contain the following sections and subsections in detail.

1. Specifications for the Design
2. Pre analysis work for the design or Models used
3. Design methodology in detail
4. Design Equations
5. Experimental techniques (if any)

Apart from the aforementioned sections, you can add sections as per the requirements of the project in consultation with your guide.

3.2 Paraphrasing

When you paraphrase a written passage, you rewrite it to state the essential ideas in your own words. Because you do not quote your source word for word when paraphrasing, it is unnecessary to enclose the paraphrased material in quotation marks. However, the paraphrased material must be properly referenced because the ideas are taken from someone else whether or not the words are identical.

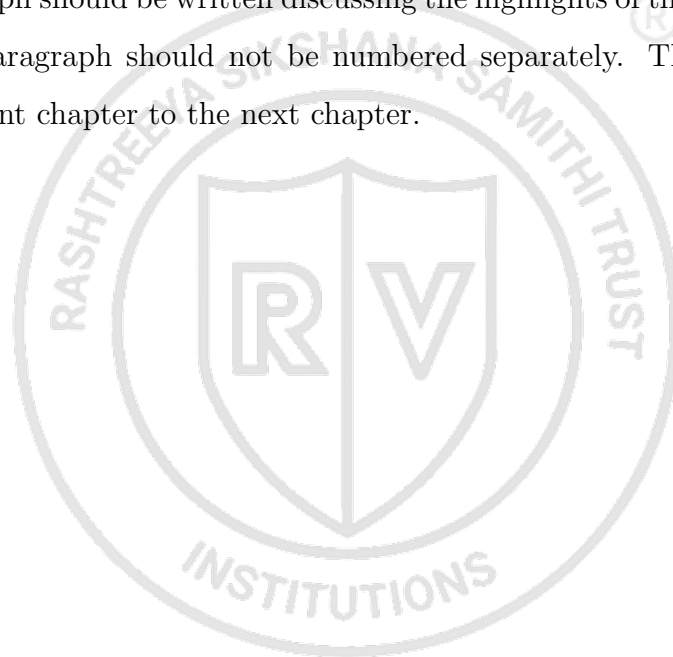
Ordinarily, the majority of the notes you take during the research phase of writing your report will paraphrase the original material. Paraphrase only the essential ideas. Strive to put original ideas into your own words without distorting them."

3.3 Quotations

When you have borrowed words, facts, or idea of any kind from someone else's work, acknowledge your debt by giving your source credit in footnote (or in running text as cited reference). Otherwise, you will be guilty of plagiarism. Also, be sure you have represented the original material honestly and accurately. Direct word to word quotations are enclosed in quotation marks.”

The chapters should not end with figures, instead bring the paragraph explaining about the figure at the end followed by a summary paragraph.

After elaborating the various sections of the chapter (From Chapter 2 onwards), a summary paragraph should be written discussing the highlights of that particular chapter. This summary paragraph should not be numbered separately. This paragraph should connect the present chapter to the next chapter.



Chapter 4
**Implementation of Pipelined Analog
to Digital converter**

CHAPTER 4

IMPLEMENTATION OF PIPELINED ANALOG TO DIGITAL CONVERTER

Every chapter should start with an introduction paragraph. This paragraph should brief about the flow of the chapter. This introduction can be limited within 4 to 5 sentences. The chapter heading should be appropriately modified (a sample heading is shown for this chapter). But don't start the introduction paragraph in the chapters like "This chapter deals with....". Instead you should bring in the highlights of the chapter in the introduction paragraph.

4.1 Contents of this chapter

This chapter should elaborate the following in detail.

1. Implementation details for hardware based projects
2. Top level Design for software based projects

You can add sections and sub sections to elaborate your project work done.

The chapters should not end with figures, instead bring the paragraph explaining about the figure at the end followed by a summary paragraph.

After elaborating the various sections of the chapter (From Chapter 2 onwards), a summary paragraph should be written discussing the highlights of that particular chapter. This summary paragraph should not be numbered separately. This paragraph should connect the present chapter to the next chapter.



Chapter 5

Results & Discussions

CHAPTER 5

RESULTS & DISCUSSIONS

Every chapter should start with an introduction paragraph. This paragraph should brief about the flow of the chapter. This introduction can be limited within 4 to 5 sentences. The chapter heading should be appropriately modified (a sample heading is shown for this chapter). But don't start the introduction paragraph in the chapters like "This chapter deals with...". Instead you should bring in the highlights of the chapter in the introduction paragraph.

5.1 Contents of this chapter

All the results obtained for your objectives should be discussed in this chapter. This chapter should contain the following sections as per the project.

1. Simulation results
2. Experimental results
3. Performance Comparison
4. Inferences drawn from the results obtained

All the figures should be properly explained by bringing the scenarios of the design done in the project. A detailed discussion of results obtained should be done in this chapter.

5.2 Tables in thesis

- All Table Caption should be in Sentence Case, TNR 10 Pt. It should be of the Format:
 - Table 1.1 Results of the experiment ... (Centered)
- It should be cited as Table 1.1.
- Caption should appear above the Table.
- Table Header and the entries should be of Font TNR 10 Pt, Justified.
- For wider Table, the page orientation can be Landscape.

- For Larger Table, it can run to pages and the header should be repeated for each page of the Table.
- Table must be adjusted to fit in the page and no single row is left out for a new page.

Sample Table 5.1 is given below for your reference,

Table 5.1: Country List

Country Name or Area Name	ISO ALPHA 2 Code	ISO ALPHA 3 Code	ISO numeric Code
Afghanistan	AF	AFG	004
Aland Islands	AX	ALA	248
Albania	AL	ALB	008
Algeria	DZ	DZA	012
American Samoa	AS	ASM	016
Andorra	AD	AND	020
Angola	AO	AGO	024

5.3 Math equation in thesis

All equation should be written using equation editor or using an equivalent tool.

- Equations should be numbered as : 1.1, 1.2 ...
- Equation should be Centered, 12 Pt, TNR.
- Equation number should be right Justified
- It should be cited as Eqn. 1.1.
- If the sentence starts by citing an equation, then it should be written as Equation 1.1 For example, Equation 5.1 states the Pythagoras theorem.

For example in Eqn. 5.1, The well known Pythagorean theorem $x^2 + y^2 = z^2$ was proved to be invalid for other exponents. Meaning the next equation has no integer solutions:

$$x^n + y^n = z^n \quad (5.1)$$

The mass-energy equivalence is described by the famous equation in Eqn. 5.2

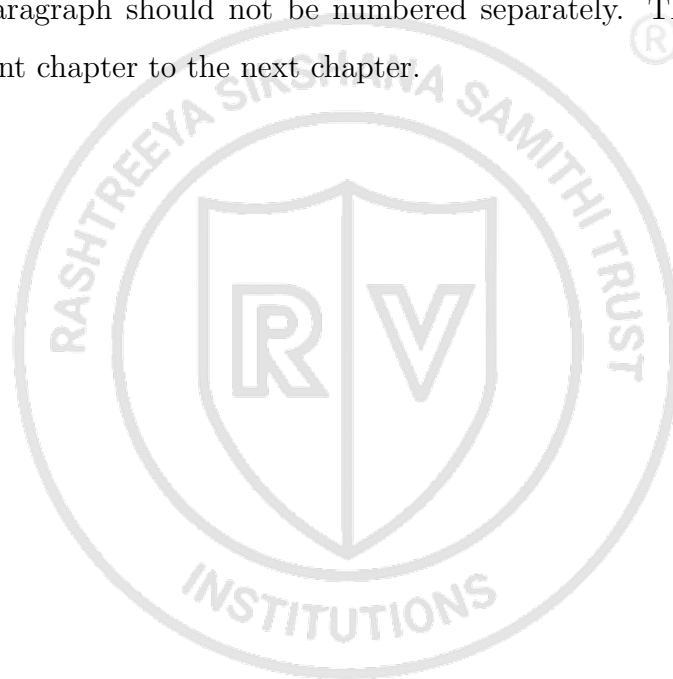
$$E = mc^2 \quad (5.2)$$

discovered in 1905 by Albert Einstein.

5.4 Interpretation of results

The chapters should not end with figures, instead bring the paragraph explaining about the figure at the end followed by a summary paragraph.

After elaborating the various sections of the chapter (From Chapter 2 onwards), a summary paragraph should be written discussing the highlights of that particular chapter. This summary paragraph should not be numbered separately. This paragraph should connect the present chapter to the next chapter.





Chapter 6

Conclusion and Future Scope

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

Every chapter should start with an introduction paragraph. This paragraph should brief about the flow of the chapter. This introduction can be limited within 4 to 5 sentences. The chapter heading should be appropriately modified (a sample heading is shown for this chapter). But don't start the introduction paragraph in the chapters like "This chapter deals with...". Instead you should bring in the highlights of the chapter in the introduction paragraph.

6.1 Conclusion

This chapter should not contain an introduction paragraph like other chapters. You can directly write conclusion of the work done under this section. Typically this section can have 3 to 4 paragraphs.

First paragraph should bring in the scenario of the project and every objective should be explained here.

Second paragraph should say how the objectives are implemented and achieved.

Last paragraph should draw the conclusions from each objective with quantitative results, performance improvement etc.

6.2 Future Scope

Briefly discuss the constraints and limitations of the project and state the possibilities of extending the work in future.

6.3 Learning Outcomes of the Project

- List the learning outcomes here
- List a minimum of 5 learning outcomes



Appendix A

Code

APPENDIX A

CODE

A.1 First Appendix

You can use `tcbllisting` for creating the code snippets. The following example illustrates how one can customize the `tcbllisting` to achieve the `tcl` script. Similarly, one can use it for other programming language listing, including HDL.

```
# Since our design has a clock with name clk,
## specify that name under [get_port ]
create_clock -period 40 -waveform {0 20} [get_ports clk]

# Setting a 'delay' on the clock:
set_clock_latency 0.3 clk

# Setting up constraints on your I/P and O/P pins
set_input_delay 2.0 -clock clk [all_inputs]
set_output_delay 1.65 -clock clk [all_outputs]

# Set realistic 'loads' on each output pin
set_load 0.1 [all_outputs]

# Set 'maximum' fanin and fan-out for the input and output pins
set_max_fanout 1 [all_inputs]
set_fanout_load 8 [all_outputs]
```

BIBLIOGRAPHY

- [1] IMTF Compliance Technologies, *When trade becomes a cover: Unpacking trade-based money laundering*, 2024. [Online]. Available: <https://imtf.com/blog/when-trade-becomes-a-cover-unpacking-trade-based-money-laundering/>
- [2] Financial Action Task Force and Egmont Group, “Trade-based money laundering: Trends and developments,” FATF, Paris, France, Tech. Rep., 2020. [Online]. Available: <https://fatf-gafi.org>
- [3] United Nations Office on Drugs and Crime, *Money laundering and Illicit financial flows overview*, 2024. [Online]. Available: <https://unodc.org>
- [4] Europol, *Criminal finances, money laundering and asset recovery*, 2024. [Online]. Available: <https://europa.eu>
- [5] J. Saenz and J. Lewer, “Quantifying illicit financial flows and trade misinvoicing in developing economies,” *Journal of Financial Crime Research*, vol. 18, no. 2, pp. 112–128, 2024.
- [6] B. Razavi, *Design of Analog CMOS Integrated Circuits*. McGraw-Hill Education, 2000, ISBN: 978-0072380323.